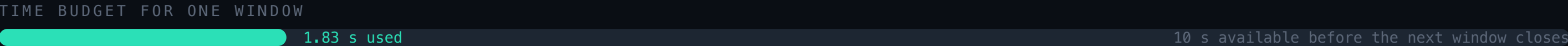
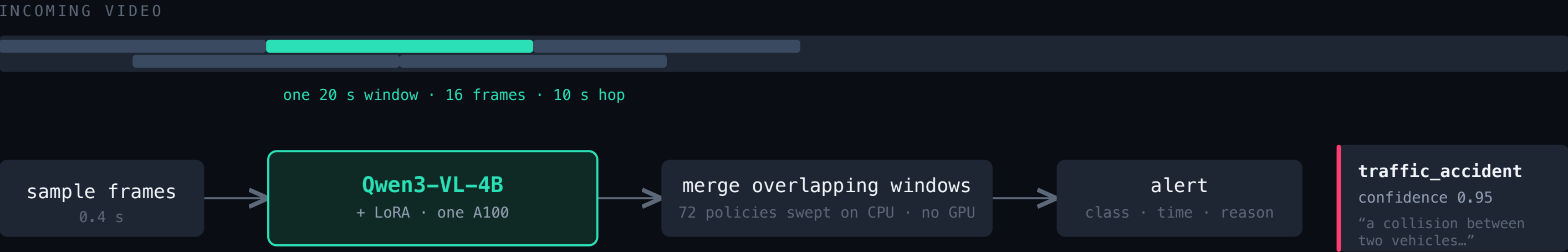


WHAT WE BUILT

A 4B model watches the video and raises the alert itself.

Real-time video anomaly detection · 11 classes
Qwen3-VL-4B-Instruct · one A100-40GB
34 public test videos · 56 minutes of footage

Every 10 seconds a new 20-second window closes. 16 frames go to the model; it answers with a class, a confidence and a sentence of reasoning.



75 / 100

marks had no training data
Train ships 5–30 s clips with one event. D2 and D3 are 4–10 minute videos with several. We synthesised 360 of them.

5.5x

faster than real time
1.83 s to score a window that arrives every 10 s. Still 2.8x at the 95th percentile, so it holds on the tail.

4B

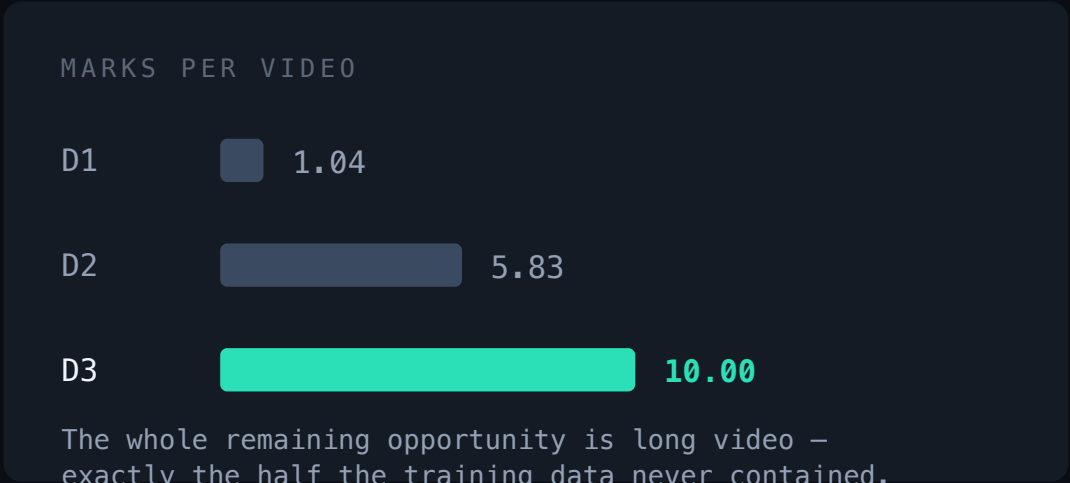
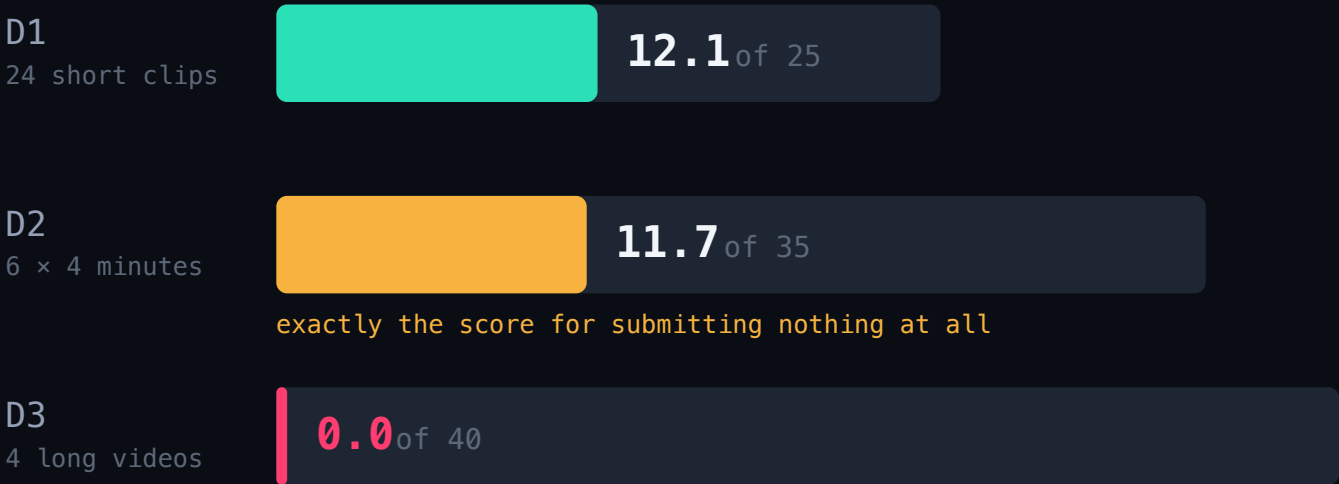
parameters, one GPU
The 8B is statistically indistinguishable on this test, so we ship the smaller one. Nothing hosted is in the runtime path.

Demo: a dashboard runs this live — press a button, one window goes to the A100, the alert comes back in about nine seconds.

Short clips work. Long video does not — and that is where the marks are.

First scored upload · zero-shot, no fine-tuning yet
23.8 / 100
24 videos, 20 positives → ±1 detection ≈ 5 pp of recall

Marks earned against marks available. One D3 video is worth ten D1 videos.



12 → 3

false alarms, for free
Keeping only the top answer per clip cut them 4x with no loss of recall, on CPU. The model's own confidence turned out to be useless as a filter.

7

silent failures, 0 crashes
Every one would have read as “the model isn't very good” — and the fix for the fifth caused the seventh. So we validate intermediate artifacts, not outputs.

0.556

AUC where chance is 0.500
The ego-motion gate every fixed-camera paper implies. We tested it against a permutation null: it does not work. A day of GPU saved.